Full Stack Development with MERN

Project Documentation format

1. Introduction

- **Project Title:**
  Enforcing Mandatory Fields Using UI Policies and Migrating Changes with Update Sets

- **Team Leader:**
  Gokul M

- **Team Members:**
  Yogesh Kumar R , Balaji B , Yokesh J

2. Project Overview

Purpose

The purpose of this project is to ensure accurate data entry by enforcing mandatory fields using UI Policies and to enable safe and controlled migration of configuration changes using Update Sets in ServiceNow.

Features

- Dynamic enforcement of mandatory fields
- Conditional field visibility and read-only control
- Real-time form validation
- Automatic capture of configuration changes using Update Sets
- Safe migration across environments (Dev → Test → Prod)
- Error-free deployment and tracking

3. Architecture

Frontend (User Interface)

- ServiceNow Forms (Incident, etc.)

- UI Policies for dynamic behavior
- Client Scripts for validation

Backend (Application Logic)

- ServiceNow platform logic
- UI Policy Engine
- Update Set Management system

Database

- ServiceNow Database (MySQL-based)
- Stores records, configurations, and logs

4. Setup Instructions

Prerequisites

- ServiceNow Developer Instance
- Web browser (Chrome/Edge)
- Basic knowledge of ServiceNow

Installation / Setup

1. Log in to ServiceNow instance
2. Navigate to UI Policies module
3. Create UI Policies and define conditions
4. Configure forms and fields
5. Create and manage Update Sets
6. Test functionality

5. Folder Structure (Conceptual)

Since this is a ServiceNow-based project, structure is module-based:

- UI Policies → Field behavior rules
- Forms & Tables → Data structure
- Update Sets → Configuration tracking
- Client Scripts → Additional validation logic

6. Running the Application

- Open ServiceNow instance in browser
- Navigate to relevant module (e.g., Incident)
- Create or edit records
- UI Policies will automatically enforce rules

7. Configuration / API Documentation

- UI Policies define:
    - Mandatory fields
    - Visibility rules
    - Read-only conditions
- **Update Sets:**
    - Capture changes
    - Enable migration
    - (Optional) ServiceNow REST APIs can be used for integration

8. Authentication & Security

- Role-Based Access Control (RBAC)
- Access Control Lists (ACLs)
- Secure login via ServiceNow authentication
- HTTPS encryption for data protection

9. User Interface

- Dynamic ServiceNow forms
- Fields automatically become:
    - Mandatory
    - Visible/Hidden
    - Read-only
- Improved user experience with guided input

10. Testing

- Functional testing of UI Policies
- Validation testing for mandatory fields
- Migration testing using Update Sets
- User Acceptance Testing (UAT)

11. Screenshots / Demo

- UI Policy configuration screen
- Incident form with mandatory fields
- Update Set creation and migration
  (You can attach screenshots here)

12. Known Issues

- Complex UI Policy conditions may require careful testing
- Dependency issues during migration if not validated properly

13. Future Enhancements

- Add AI-based validation for smarter field enforcement
- Automated conflict detection before migration
- Dashboard for monitoring Update Set status
- Integration with other enterprise systems